

Sentrais Core OS & Resilience Orchestration Manual

Version 2.0 – Strategic Architecture Edition

Classification: Internal | Executive | Enterprise Deployment Reference

1. Purpose

This document defines the canonical architecture, governance model, innovation framework, and deployment lifecycle for all Sentrais-powered ecosystems.

It serves four purposes:

1. Enterprise alignment across all vertical deployments
2. Investor and partner architecture clarity
3. Governance and IP boundary reinforcement
4. Training and certification standardization

Sentrais is no longer a single deployment system. It is a modular orchestration operating system powering sector-specific intelligence environments.

2. Sentrais Platform Architecture

2.1 The Platform Stack

Sentrais operates through a layered intelligence model and a verticalized edge application strategy.

Layer 1: Infrastructure & Cloud Fabric

Cloud-agnostic, multi-region architecture
Zero Trust networking model
Environment isolation: Dev | Test | Staging | Production
Disaster recovery with defined RTO/RPO

Layer 2: Sentrais Core OS

The Core OS is the protected intelligence engine.

Core components:

- Blueprint Engine
- Entity Registry
- Dependency Graph
- Evidence Ledger
- Orchestration Engine
- Resilience Metrics Engine
- Role & Identity Framework
- Governance & Change Control System

The Core OS is not client-owned. It is licensed.

Layer 3: Edge Applications

Each vertical deployment operates as an orchestration layer built on the Core OS.

EVERGAME

League and competition orchestration

eVenu

Venue certification and revenue intelligence

CiviGrid

Host city and multi-agency coordination

SearGrid

National-level SEAR orchestration and cross-jurisdiction resilience

Replaces prior SEAR Engine terminology.

Focused on federal alignment, state coordination, and multi-event continuity.

EntertainmentOS

Entertainment and live production orchestration

Concerts, festivals, broadcast, media operations

Innovation OS

Challenge orchestration, R&D lifecycle, model training, and ecosystem simulation

Each Edge Application inherits the Core OS but configures sector-specific logic, dashboards, integrations, and certification models.

3. Resilience Orchestration Framework

Sentra operationalizes resilience through a defined cycle.

3.1 The Orchestration Cycle

Readiness → Run → Response → Review → Renewal

At the center is the Metrics Engine.

Key measurable outputs:

- Readiness Score
- Incident Response Time
- SLA Compliance
- Certification Status

- Learning Velocity
- Recovery Benchmark

All deployments must support this cycle.

4. The Six-Layer Intelligence Model

All deployments must adhere to the canonical intelligence stack.

4.1 Ingestion Layer

APIs

Streaming data

Manual entry

External system connectors

4.2 Blueprint & Entity Layer

Digital twin mapping

Entity registry

Dependency mapping

Operational topology

4.3 Intelligence Layer

Simulation

Anomaly detection

Predictive modeling

Scenario stress testing

4.4 Evidence Layer

Immutable audit log

Decision traceability

Compliance validation

Post-event reconstruction

4.5 Orchestration Layer

- Task sequencing
- Role routing
- Escalation triggers
- Cross-system coordination

4.6 Security & Governance Layer

- RBAC
- MFA
- Zero Trust enforcement
- Compliance mapping
- Change control

This layered model is mandatory for certification.

5. Innovation Framework Integration

Innovation OS and SearGrid deployments must align to the Sentrais Innovation Framework.

5.1 The Innovation Orchestration Framework

Sentrais applies a structured five-phase model across ecosystems.

Phase 1: Discover

- Map ecosystem entities
- Capture constraints
- Identify systemic dependencies

Phase 2: Diagnose

- Expose operational blind spots
- Identify friction and fragmentation
- Quantify risk and resilience gaps

Phase 3: Design

Architect the Blueprint Model
Define interoperability schema
Establish governance rules

Phase 4: Deploy

Activate orchestration logic
Integrate systems
Validate intelligence outputs

Phase 5: Debrief

Run simulations
Capture evidence
Refine model
Renew certification

This framework governs Innovation OS and informs every vertical deployment.

6. Deployment Lifecycle 2.0

Discovery
Design
Build
Integrate
Secure
Deploy
Certify
Sustain
Renew

6.1 Sustain & Govern Phase

Quarterly readiness reviews
Continuous SLA validation
Model drift detection
Security audits
Evidence ledger monitoring

6.2 Annual Recertification

Mandatory for:

- SearGrid deployments
- CiviGrid host cities
- eVenu certified venues
- EntertainmentOS major productions

Certification is evidence-based, not attestation-based.

7. AI Governance & Responsible Model Oversight

All intelligence models must comply with:

Human-in-the-loop review
Explainability logging
Bias monitoring
Model drift detection
Simulation validation prior to production release

AI is treated as a controlled subsystem within the Orchestration Layer.

8. Governance & IP Boundary Model

8.1 Decision Rights

Client retains:
Operational decisions
Role assignments
Local configuration

Sentrais retains:

- Core OS IP
- Evidence Ledger structure
- Orchestration logic engine
- Blueprint architecture methodology

Edge configuration is licensed per deployment agreement.

9. Integration Framework

Integrations are classified as:

- Core Integrations
- Operational Integrations
- Analytic Integrations

All integrations require:

- Interface Control Document
- Security mapping
- Latency validation
- Failure protocol
- SLA assignment

Multi-region resilience is required for SearGrid and national deployments.

10. Certification & Evidence Standard

Certification requires:

- Completed simulation
- Validated dashboards
- Security verification
- Integration validation
- Evidence ledger review

No deployment is considered enterprise-grade without certification artifacts.

11. Strategic Positioning

Sentra is not a dashboard product.
It is an orchestration operating system.

It enables:

League-scale operations
National resilience coordination
Entertainment ecosystem synchronization
Innovation lifecycle governance

The value is not visualization.
The value is coordination with proof.

12. Version Control

This document supersedes Version 1.1.
All future deployments must align to Version 2.0 architecture and naming conventions.